

Experiment 4: polyalphabetic substitution

```
#include <stdio.h>

#include <string.h>

#include <ctype.h>

int main()
{
    char text[100], key[100];

    int i, j = 0, k;

    printf("Enter plaintext: ");
    scanf("%[^\n]", text);

    printf("Enter key: ");
    scanf("%s", key);

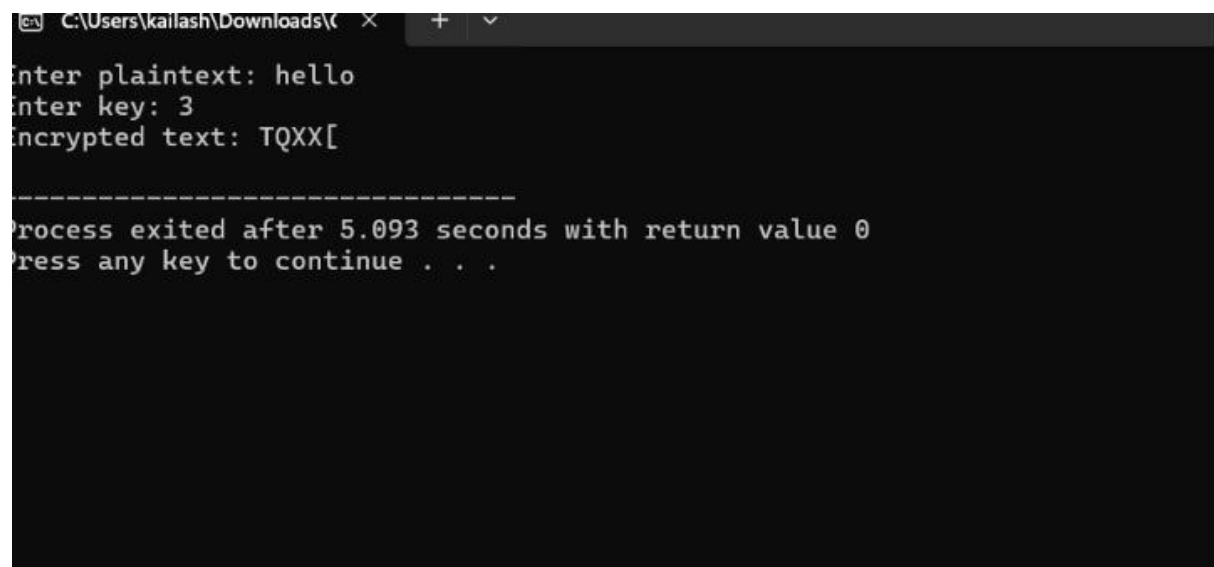
    printf("Encrypted text: ");

    for (i = 0; text[i] != '\0'; i++)
    {
        if (isalpha(text[i]))
        {
            k = tolower(key[j % strlen(key)]) - 'a';

            if (isupper(text[i]))
                printf("%c", (text[i] - 'A' + k) % 26 + 'A');
            else
                printf("%c", (text[i] - 'a' + k) % 26 + 'a');
```

```
        j++;  
    }  
    else  
    {  
        printf("%c", text[i]);  
    }  
}  
  
printf("\n");  
return 0;  
}
```

Output:



```
C:\Users\kailash\Downloads\ > .\program.exe  
Enter plaintext: hello  
Enter key: 3  
Encrypted text: TQXX[  
-----  
Process exited after 5.093 seconds with return value 0  
Press any key to continue . . .
```